4 **(a)** Define *electric potential* at a point.

work done per unit positive charge by an external force [B1] in moving a small test charge from infinity to that point in the electric field [B1] (without a change in its kinetic energy).

.... [2]

(b) Two charged solid spheres A and B are situated in a vacuum. Their centres are separated by a distance of 30.0 cm, as shown in Fig. 4.1. The diagram is not drawn to scale.

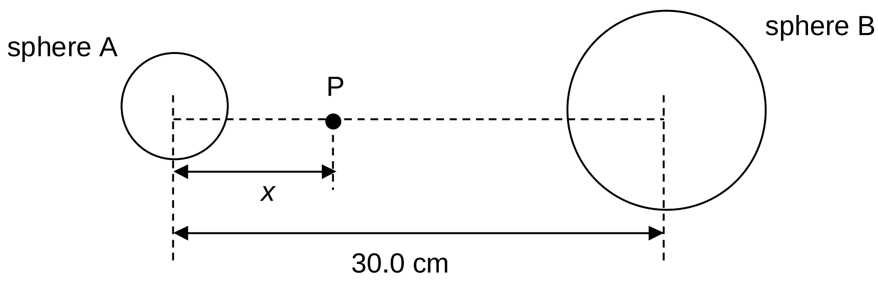

Fig. 4.1

The variation with distance x of the electric potential V_A and V_B due to sphere A and B independently is shown in Fig. 4.2.

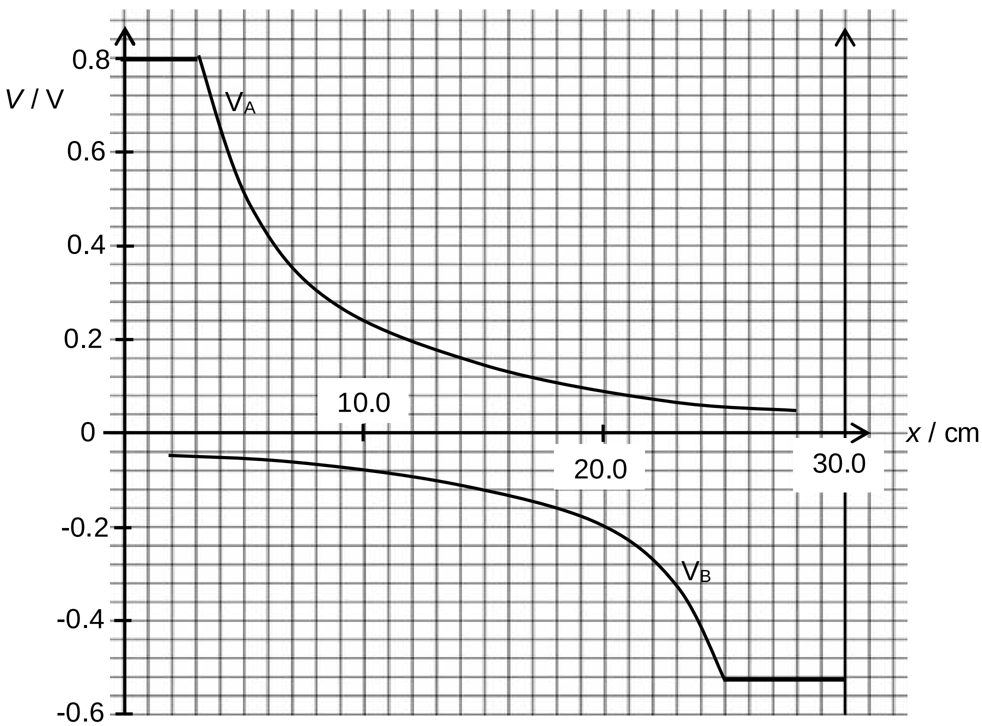

Fig. 4.2

For Examiner's Use

(i) Using Fig. 4.2, state the radius of both spheres.

Sphere A: 3.0 cm
Sphere B: 5.0 cm

radius of sphere A = ……………… cm

radius of sphere B = ………………. cm [1]

(ii) State and explain the signs of both spheres.

Since the potential of sphere A is positive and that of sphere B is negative, sphere A is positively charged while sphere B is negatively charged.

1 mark – opposite charges

(Cannot accept using gradient to determine direction of E field since no indication of which direction of *x* is positive.)

…… [2]

(iii) Point P is at a distance x = 10.0 cm.
An alpha particle has kinetic energy E_K when at infinity.

Use Fig. 4.2 to determine the minimum value of E_K such that the alpha particle may travel from infinity to point P.

Resultant potential (10 cm from sphere A) = 0.24 + (-0.08) = 0.16 V [B1]

By COE,

Loss in KE = Gain in EPE

$E_K - 0 = 0.16\ (2 \times 1.6 \times 10^{-19}) - 0$ [M1 – 2q, qΔV, COE]

$E_K = 5.12 \times 10^{-20}$ J [A1]

If B1 correct and M1 wrong, 1 mark.
If B1 wrong and M1 correct, 1 mark.

E_K = ……………………….. J [3]

For Examiner's Use

5 An ideal transformer is connected to a sinusoidal a.c. supply, as shown in Fig. 5.1. The primary